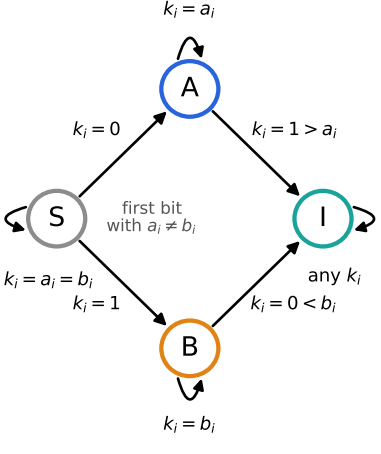
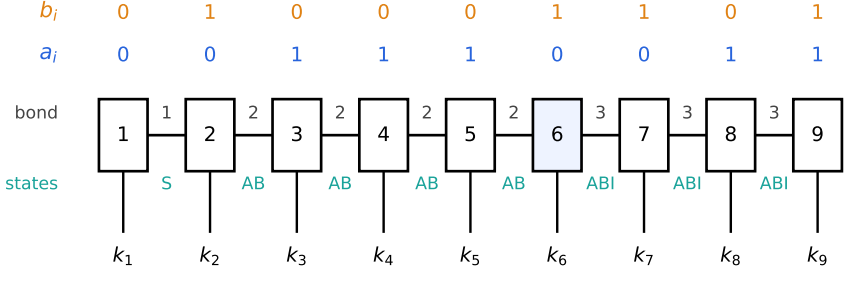
(a) the comparator for $a \le k \le b$



(b) the train of the mask $115 \le k \le 141$ ($a = 001110011$, $b = 010001101$)



(c) the core of bit 6 ($a_6 = 0$, $b_6 = 1$): rows the state before, columns the state after

$k_6 = 0$:

	A	B	I
A	1	0	0
B	0	0	1

$k_6 = 1$:

	A	B	I
A	0	0	1
B	0	1	0